GTRY154 Magnetic Keyboard Sensor and LED Driver

Chip Datasheet

V1.1

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1. General Description

GTRY154 is a low power chip for magnetic backlit keyboard application, which integrates a Hall sensor for magnetic switch and a RGBW LED Driver into one chip. The Hall sensor senses the distance of the magnet in the switch, and digitizes it into a digital sample value by the internal magnetic processing unit, and the digital sample value can be transmitted to the MCU through the 3-wire serial port.

With the 5rd generation Hall detecting technology, GTRY154 can realize up to 1.1 Gauss resolution and 8KHz sampling rate for full size keyboard. GTRY154 achieves a sampling rate of 104-key keyboard less than 0.1ms. It uses constant current driver to drive RGBW LED, and this RGBW four-channel constant current value can be adjusted separately. Combined with the 13bit PWM, a very smooth and fine- tuned true color can be achieved.

Multiple 2-in-1 chips can be connected in series by daisy chain, and each chip parameter on the daisy chain can be separately configured through the 3-wire serial port, such as RGBW current and PWM parameters.

2. Fearures

- **Communication**
  - 3-wire serial port
  - Support daisy chain serial control
  - The magnetic induced voltage is quantized into a digital sample values
  - Supprot 8KHz reporting rate
- **PWM**
  - Supports 4 channel 13-bit constant current open-drain PWM output
  - Supports brightness adjustment with a 15 bit PWM cycle
  - Individual 17 coarse DC current steps,they are 0.6 / 0.9 / 1.2 / 1.8 / 2.4 / 2.7 / 3.0 / 3.6 / 4.2 / 4.5 / 4.8 / 5.4 / 6.0 / 6.3 / 7.2 / 8.1 / 9.0mA
  - 256 fine DC current adjustment for Individual Corase current
- **Magnetic axis adaptability**
  - Up to 1.1 Gauss resolution
  - Input signal range supports±(20~500)mT configurable
  - Supports temperature compensation, configurable from -600 to 2500ppm/℃
- **Power supply**
  - Power supply voltage range of VDD：3.1~5.5V
  - Supprot two sleep mode : Sleep power 3uA，DeepSleep power 0.1uA
  - Supports **SPS**[smart power-saving] mode, which can reduce power consumption by 60%
- **ESD performance and temperature**
  - 2KV HBM
  - -10°C～+70°C
- **Package**
  - DFN2x2-8
- **Application**
  - USB RGB Magnetic Keyboard
  - 2.4GHz or BLE Magnetic Keyboard

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3. Functional Block Diagram

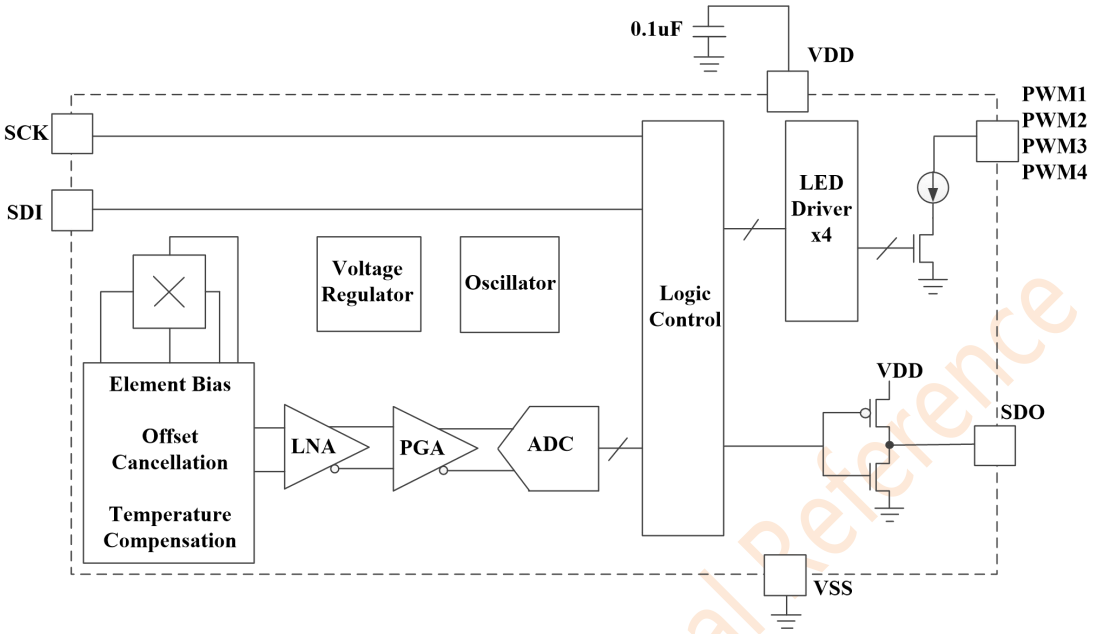


Fig 1. Functional Block

4. Pin Assignment

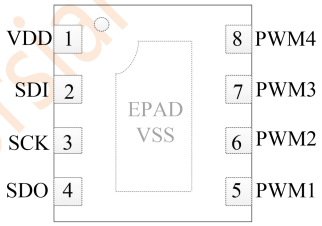


Fig 2. Top View

Table1 Pin description

Num	Name	I/O	Description	Remark
1	VDD	—	Supply voltage	
2	SDI	I	Serial data input	
3	SCK	I	Clock input	
4	SDO	O	Serial data output	
5	PWM1	O	PWM Constant current open drain output for LED4	
6	PWM2	O	PWM Constant current open drain output for LED3	
7	PWM3	O	PWM Constant current open drain output for LED2	
8	PWM4	O	PWM Constant current open drain output for LED1	
9	VSS	—	Ground EPAD	

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5. Electrical Characteristics

5.1 Absolute Maximum Ratings (at TA=25°C)

The working conditions of the chip exceed the following range, which will affect the reliability of the device.

Characteristics	Symbol	Rating	Unit
VDD Pin Voltage	V_{DD}	-0.3~5.5	V
Input voltage	V_{in}	-0.3~5.5	V
Output voltage	V_{out}	-0.3~5.5	V
Storage Temperature Range	T_{ST}	-40 ~ 85	℃
Operating Junction Temperature	T_J	115	℃
Operating Temperature Range	T_{OP}	-10 ~70	℃
Electrostatic Discharge Voltage	HBM	2.0	KV
	CDM	500	V
Steady State Life Test@85℃	-	3000	hrs
Welding Temperature	-	260 (<10s)	℃

5.2 DC Electrical Characteristics

Characteristics	Symbol	Min	Typ	Max	Unit
Supply Voltage of VDD	V_{DD}	3.1	5.0	5.5	V
Supply Current @ Active	I_{Active}		1.88	2.30	mA
Supply Current @ Standby	$I_{Standby}$		0.25	0.30	mA
Supply Current @ Sleep	I_{Sleep}		3	5	uA
Supply Current @ Deep Sleep	$I_{DeepSleep}$		0.1	0.5	uA
PWM Constant Current	I_{PWM}	0.01		9.0	mA
Current sink headroom voltage of LED Driver	V_{HR}		0.50	0.60	V
Low level input voltage	V_{IL}			0.2*V_{DD}	V
High level input voltage	V_{IH}	0.5*V_{DD}			V
Low level output voltage	V_{OL}			0.2*V_{DD}	V
High level output voltage	V_{OH}	0.8*V_{DD}			V
SCK/SDI Input Capacitor	Cin			5	pF
Quiescent Output Error	QOE			±10	%
Input magnetic field range	I_m	-500		500	mT

Note :

1) VDD = 5.0V, TA = +25°C, unless otherwise noted;
2) When I_{Active}, Hall sensor & ADC enable;
3) when $I_{Standby}$, Hall sensor & ADC disable, OSC & LDO enable; For applications that only drive LEDs without considering button detection.
4) when I_{Sleep}, Hall sensor、ADC、OSC、LDO disable;
5) when $I_{DeepSleep}$, Turn off the low-voltage reset function;
6) I_{PWM} is LED constant current and it can be adjusted within the range of 0.01~9.0mA .

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5.3 SPI Electrical Characteristics

IC Support 4-wire slave SPI，Supports full duplex data transmission with a data bit width of 8 bits。

Table2. SPI Interface Description

Num	SPISignal name	IC Signal name	IC Pin direction	Description
1	SPI_SS	-	-	Default enable, no chip selection signal
2	SPI_CLK	SCK	INPUT	SPI Clock
3	SPI_MOSI	SDI	INPUT	SP MOSI
4	SPI_MISO	SDO	OUTPUT	SPI MISO

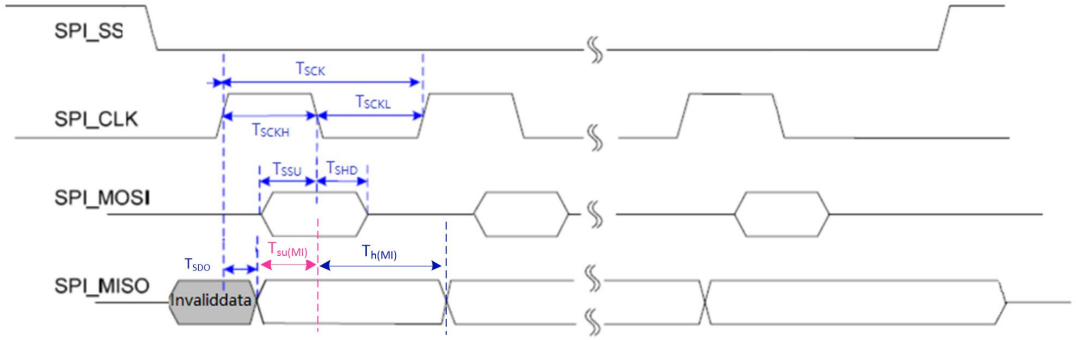


Fig 3. SPI Timing Diagram

Table3. SPI timing parameters

Symbol	Min	Max	Unit	Description
T_{SCKH}	27.5		ns	SPI_CLK High Level time
T_{SCKL}	27.5		ns	SPI_CLK Low Level time
T_{SCK}	55	1000	ns	SPI_CLK frequency range is 1.0~18MHz
T_{SSU}	10		ns	SPI_MOSI Setup time
T_{SHD}	10		ns	SPI_MOSI Hold time
T_{SDO}	20		ns	The time required for SDO to output valid data
$T_{su(MI)}$	10		ns	The setup time required to return data on the host side
$T_{h(MI)}$	4		ns	The holdon time required to return data on the host side

Note:

1. Usually, the SPI signal provided by the MCU can meet the timing requirements. However, due to the long data transmission link, the relationship between the data SDO returned and the clock needs to be confirmed, as shown in the red label in the figure above. SPI_MISO needs to meet the $T_{su\ (MI)}$ timing requirements, otherwise it will cause SPI communication errors. Please refer to the selected MCU chip for the specific value of $T_{su\ (MI)}$.
2. The detection point of $T_{su\ (MI)}$ oscilloscope testing should be as close as possible to the MCU controller pins.

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6. Typical Application mode

Connectivity	Mode	SensorScan Rate	Latency(us)	PWM Engin	Power Consumption(mA)	Remark
Wired	Wired Mode	8KHz	250	ON	1.88	USB powered
Wireless	Active 0	8KHz	250	ON	1.30	Per key auto-shift to Active 2 after idle 1-255S register configurable
Wireless	Active 1	4KHz	500	ON	1.10	Per key auto-shift to Active 2 after idle 1-255S register configurable
Wireless	Active 2	500Hz	2000	ON	0.39	Auto-shift to Active 0 or Active 1 when key depressed
Wireless	Active 3	250Hz	4ms	OFF	0.06	MCU controlled, PWM ON idle 5minutes
Wired/wireless	Idle mode	4Hz	-	OFF	0.004	MCU controlled, idle 20 minutes or USB suspend

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7. Communication Protocol

This GTRY154 uses only one SCK line to implement the communication protocol, and recognizes the instruction by calculating the high-level time of SCK and performs the corresponding action. Since SCK is also a register shift clock, in order to distinguish the instruction protocol from the normal read/write clock, the frequency of SCK as a clock must be higher than 1MHz. The read and write data is compatible with the SPI data format and supports only rising edge output data and falling edge sample data configuration options.

The data written or read is MSB first and LSB last.

7.1 Instruction Table

Num	Instruction Name	Description	Remark
1	HAL_CONVERT	Rade buttons	
2	W_PWM_REG	Write PWM data	
3	SLEEP/WAKEUP	Sleep/Wake-up instruction	
4	WR_REG	End read and write register instruction	
5	FREQ_Calibrate	Oscillator frequency calibrate	

7.2 Instruction for HAL_CONVERT

Hall conversion is 16bit data back mode. the conversion is shown in Figure 4:

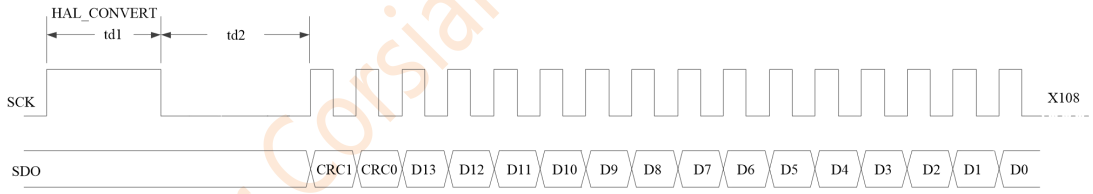


Fig 4 16bit conversion mode

The returned data is fixed 16bit,the upper two bits CRC1/CRC0 are parity bits, D13~ D0 are data bits. The relationship between the parity bits CRC1/CRC0 and the data bits is as follows, where "^" stands for "exclusive OR":

CRC0=D0^D2^D4^D6^D8^D10^D12

CRC1=D1^D3^D5^D7^D9^D11^D13

[Note]: When the return data is 0xC000 or 0x3FFF, it indicates that the chip status is abnormal or communication is interfered , and resulting in unreliable communication data,the chip needs to be reset !

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7.3 Instruction for write PWM

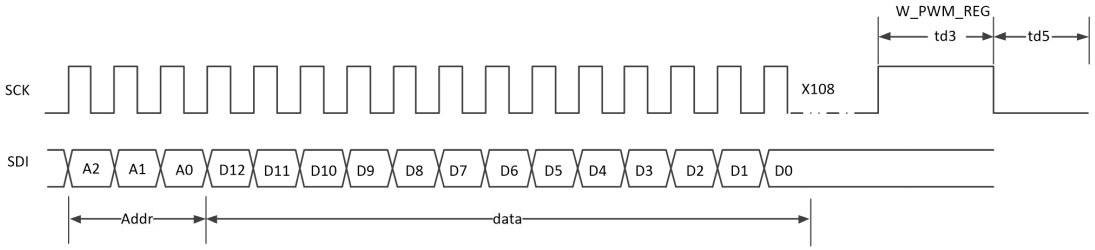


Fig 5 write PWM Instruction

The PWM register is completed by a write operation instruction. The instruction is as shown in the figure above. First, move the 3bit address and then 13bit data into the shift chain, and then send the PWM register instruction td3, wait for td5 to complete the write operation to the PWM register. The meaning of 16bit data is as follows:

Bit[15:14]: A2/A1/A0

① 3`b1xx：Write Bit[14:0] , [A1,A0,D12:D0] 15bit PWM Cycle for all LED
② 3`b011：Write Bit[12:0] , [D12:D0] PWM4 Duty Cycle for LED4
③ 3`b010：Write Bit[12:0] , [D12:D0] PWM3 Duty Cycle for LED3
④ 3`b001：Write Bit[12:0] , [D12:D0] PWM2 Duty Cycle for LED2
⑤ 3`b000：Write Bit[12:0] , [D12:D0] PWM1 Duty Cycle for LED1

7.4 Instruction for WR_REG

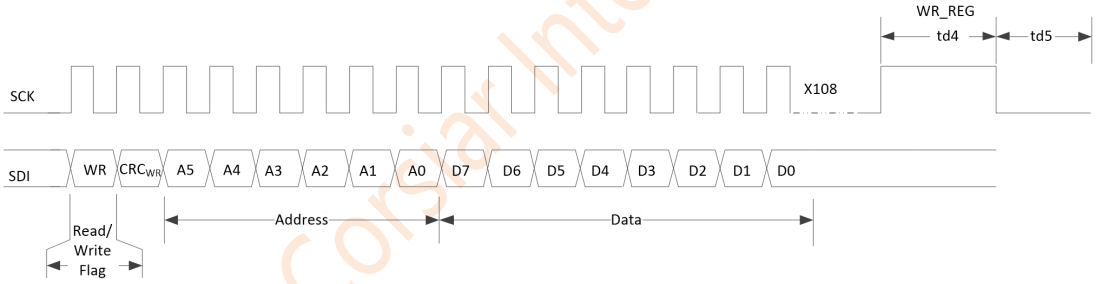


Fig6 Write Register Instruction

The operation of writing the register is as shown in Fig 6. The write operation of the register requires one instructions to be executed. First, the 1-bit read/write flag bit "1" and 1-bit CRC_{WR},and the 6-bit address are shifted into the shift chain through the SPI-like interface timing. 8-bit data, and then send and write end command to complete the write register operation.

$$CRC_{WR}=D7\text{^}D6\text{^}D5\text{^}D4\text{^}D3\text{^}D2\text{^}D1\text{^}D0$$

Note: if the value of CRC_{WR} is incorrect,the instruction is invalid.

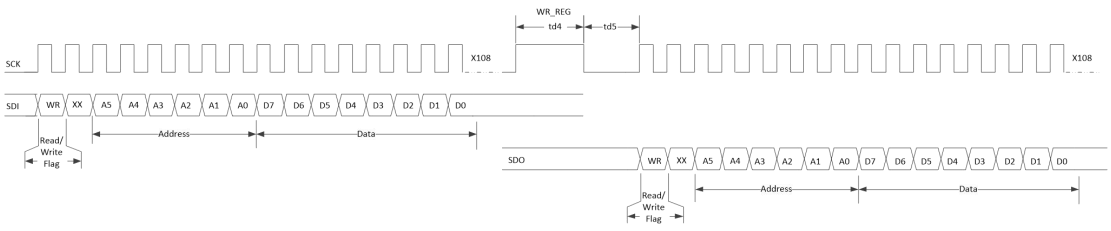


Fig 7 Read Register Instruction

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The operation of reading the register is as shown in Fig7. The completion of a read register operation also requires one instructions to complete, first write the 1-bit read/write flag bit “0”, 7-bit address, and 8-bit data through the SPI-like interface timing， and then the 8bit data can be any value. then send the end read and write register instructions, and finally read the data back from SDO through the SPI-like interface timing, read back the data is 16bit, of which the lower 8bit is valid data.

7.5 Instruction for FREQ_Calibrate

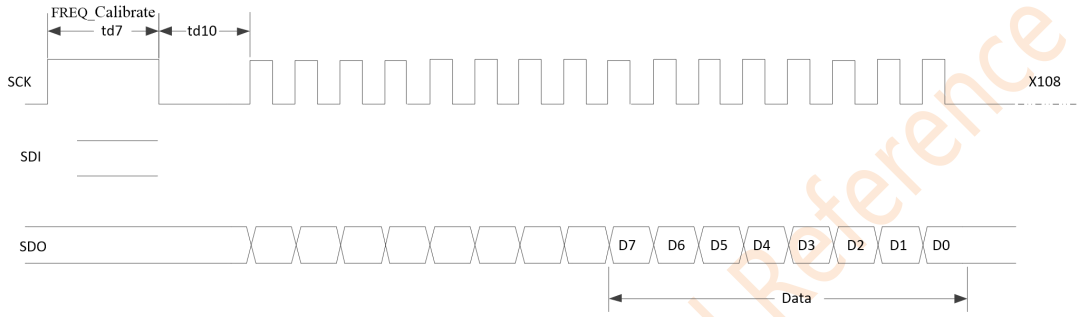


Fig 8 FREQ_Calibrate Instruction

This instruction is used to Calibration the internal oscillator frequency of the Hall chip. The instruction is as shown in Fig8. After sending the instruction, waiting for the fixed delay td10, the frequency value can be read back through the SPI-like interface timing, and the read back data is 16bit, where the lower 8 bits are valid data。

7.6 Instruction for SLEEP/WAKEUP

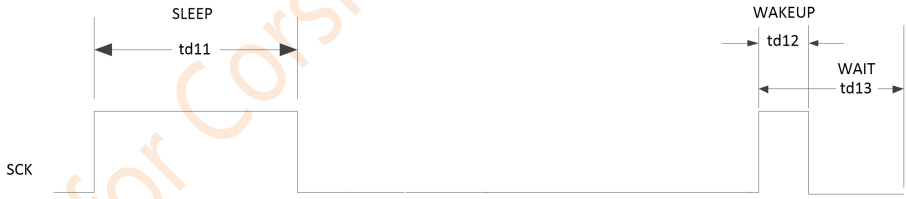


Fig 9 Sleep Instruction

The sleep instruction can put the chip into a low-power sleep state. The instruction is as shown in Fig9. After the SLEEP instruction is sent, the chip goes to sleep. During the sleep, SCK needs to remain at a low level. When SCK is pulled high and td12 is maintained, the chip will be waked up, and it can work after td13.

Note: when DeepSleepEn=1, after the SLEEP instruction is sent, the chip goes to deepsleep. About `DeepSleepEn` configure, please refer to register 0x20 for details.

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7.7 Timing table of instruction

Num	Symbol	Timing (Unit : us)	Remark
1	td1	5 ± 5%	HAL_CONVERT
2	td2	Continuous mode : >2 Single mode : >80	HAL_CONVERT Wait time
3	td3	9 ± 5%	W_PWM_REG
4	td4	24 ± 5%	WR_REG
5	td5	>2	
6	td6	-	Reserve
7	td7	128 ± 5%	FREQ_DETECT
8	td8	-	Reserve
9	td9	-	Reserve
10	td10	>2	
11	td11	15 ± 5%	SLEEP
12	td12	>2	WAKEUP
13	td13	>12	WAKEUP Wait time

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8. Register Description

The all register address is 6bit,bit width is 8bit:

Name	Addr	WR	Description	Default Value
SampleInterval_Fine	0x0A	WR	[7:0] : SamplingInterval_Fine Fine config interval time between two convert, 84~1100us configurable with a step of 4us Note: *(1) Time of SampleInterval_Fine= (SampleInterval_Fine*4+80) us* *(2) Unable to configure 0x0*	0x05
SampleInterval_Coarse	0x0B	WR	[7:0] : SamplingInterval_Coarse Coarse config interval time between two convert, 0~255ms configurable with a step of 1ms Note:Final Sampling interval as follow *(1) Normal mode:* Sampling Interval =SampleInterval_Fine+SampleInterval_Coarse *(2) SPS Mode:* Sampling Interval when key actuated =SampleInterval_Fine Sampling Interval when key released =SampleInterval_Fine+SampleInterval_Coarse	0x00
Test_Reg0	0x0C	WR	Write fix value 0xB5	0xB5
Test_Reg1	0x0D	WR	Write fix value 0x02	0x02
KeyReleased_Timeout	0x0E	WR	[7:0]: KeyReleased_Timeout In smart power-saving mode, when key released time exceeds 'KeyReleased_Timeout', the chip enters power-saving mode, and the 'KeyReleased_Timeout' can be configured from 1 to 255 seconds, with a step of 1 second	0x1F
SmartPower Save0	0x0F	WR	[7:0]: Low 8bit of the 'key released threshold' @ sps mode	0x00
SmartPower Save1	0x10	WR	[7]:SPS_Polarity smart power-saving mode polarity configuration, 1: 'ADC value' less than the 'key released threshold' enters power-saving mode	0x10

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Name	Addr	WR	Description	Default Value
			0: 'ADC value' greater than the "key released threshold'' enters power-saving mode [6]: Write fix value 0 [5:0]: Hihg 6bit of the 'key released threshold' @ sps mode	
Test_Reg2	0x11	WR	Write fix value 0x00	0x00
Test_Reg3	0x12	WR	Write fix value 0x47	0x47
Filter_Config	0x13	WR	[7:4]: Write fix value 0 [3:0]: 0xF: f-x32 ,Moving average of 32 datas 0xB: f-x16 ,Moving average of 16 datas 0x7: f-x8 ,Moving average of 8 datas 0x3: f-x4 ,Moving average of 4 datas 0x2: f-x2 ,Moving average of 2 datas 0x1: fb, For continue conversion mode 0x0: fa, For single conversion mode *[Note: 'Filter_Config' register must config 0x0 @ single conversion mode]*	0x00
LNA_Config0	0x14	WR	[7:0]: 0x41 : 128x gain 0x35 : 64x gain 0x25 : 32x gain 0x1D : 16x gain 0x0D : 8x gain other : Reserve	0x24
Test_Config0	0x15	WR	Write fix value 0x44	0x44
PGA_Config0	0x16	WR	[7:4]: Write fix value 0 [3:0]: PGA_Gain other : Reserve 9 : 24.0x gain 8 : 22.8x gain 7 : 21.6x gain 6 : 20.4x gain 5 : 19.2x gain 4 : 18.0x gain 3 : 16.8x gain 2 : 15.6x gain 1 : 14.4x gain 0 : 13.2x gain	0x05
Test_Reg4	0x17	WR	Write fix value 0x7C	0x7C
ModeConfig0	0x18	WR	[7]: EN_LED, LED enable config 1: Enable	0x0A

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Name	Addr	WR	Description	Default Value
			0: Disable [6] : Write fix value 0 [5]: Write fix value 0 [4]: Write fix value 0 [3]: Write fix value 1 [2]: Convert mode config 1: Continue conversion mode 0: Single conversion mode **[Note：'Filter_Config' register must config 0x0 @SingleMode]** [1]: Write fix value 1 [0]: smart power-saving mode enabled @ 1	
LED_Current_1	0x19	RW	LED1 DC Current config value K $I = K * (m * I_{max0}/256)$	0xFF
LED_Current_2	0x1A	RW	LED2 DC Current config value K $I = K * (m * I_{max1}/256)$	0xFF
LED_Current_3	0x1B	RW	LED3 DC Current config value K $I = K * (m * I_{max2}/256)$	0xFF
LED_Current_4	0x1C	RW	LED4 DC Current config value K $I = K * (m * I_{max3}/256)$	0xFF
LED_MaxCur1_2	0x1D	RW	[7:4]: LED2 I_{max2} config [3:0]: LED1 I_{max1} config --- I_{max} 4bit configure details as follows : 0x1: 0.6mA 0x2: 1.2mA 0x3/4: 1.8mA 0x5/8: 2.4mA [Default] 0x6/9: 3.0mA 0x7/A: 3.6mA 0xB/C: 4.2mA 0xD: 4.8mA 0xE: 5.4mA 0xF: 6.0mA	0x88
LED_MaxCur3_4	0x1E	RW	[7:4]: LED4 I_{max4} config [3:0]: LED3 I_{max3} config --- I_{max} 4bit configure details as follows : 0x1: 0.6mA 0x2: 1.2mA	0x88

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Name	Addr	WR	Description	Default Value
			0x3/4：1.8mA 0x5/8：2.4mA [Default] 0x6/9：3.0mA 0x7/A：3.6mA 0xB/C：4.2mA 0xD：4.8mA 0xE：5.4mA 0xF：6.0mA	
LED_SDO_ Driver	0x1F	WR	[7] : LED_Multiplier : m 1：m=1.5 0：m=1.0 [6:4]: Write fix value 0 [2:0]: SDO IO Driver current config 3'b000 : 2mA 3'b001 : 4mA 3'b011 : 8mA 3'b111 : 12mA	0x03
DeepSleepEn	0x20	WR	[7:4]: Write fix value 0 [3:0]: Deepsleep enable 0xA: Enable Others: Disable	0x00
Test_Reg5	0x21	WR	Write fix value 0x88	0x88
Temp Compensation	0x27	WR	[7:3]: Write fix value 0 [2:0]: Temperature compensation 0 : - 650ppm/℃ 1 : - 200ppm/℃ 2 : 250ppm/℃ 3 : 700ppm/℃ 4 : 1150ppm/℃ 5 : 1600ppm/℃ [Recommend] 6 : 2050ppm/℃ 7 : 2500ppm/℃	0x00

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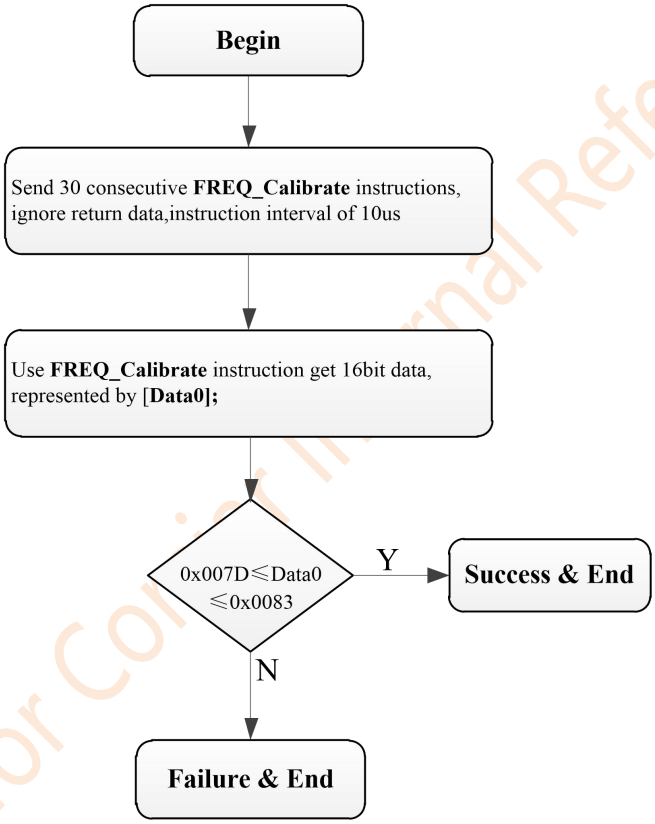
9. Initialization Process

After the IC is powered on, perform the following operations in sequence:

Step1: Correct the frequency to ensure that the chips on the daisy chain are kept in sync;

Step2: Configure the initialization parameters to ensure that the chip enters the normal working state.

The first step after power-on is to perform frequency correction to ensure that the chip on the link is synchronized. The calibration scheme is as shown in the figure below. If the calibration fails, it indicates that the chip is abnormal, and the return value can be used to determine the position of the abnormal chip in the daisy chain.



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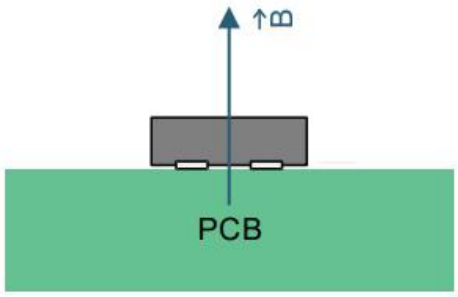
10. Low power configuration

Low power mode and wake-up need to follow the steps below:

1) Executed the SLEEP instruction to sleep;
2) Execute the WAKEUP instruction to wake up the chip;

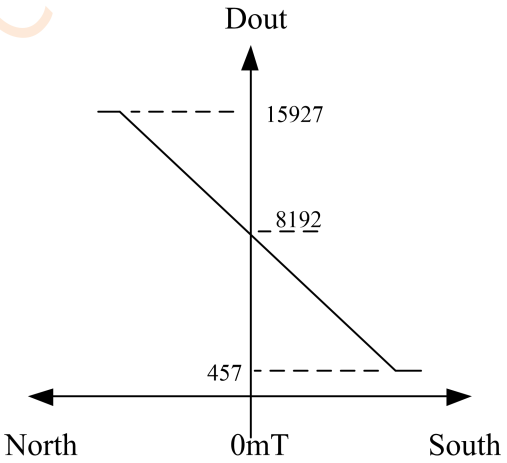
11. Magnetic Flux Direction

The device is sensitive to the magnetic field component that is perpendicular to the top of the package,direction of Sensitivity as shown in follow fig:



12. Correspondence between output and magnetic field

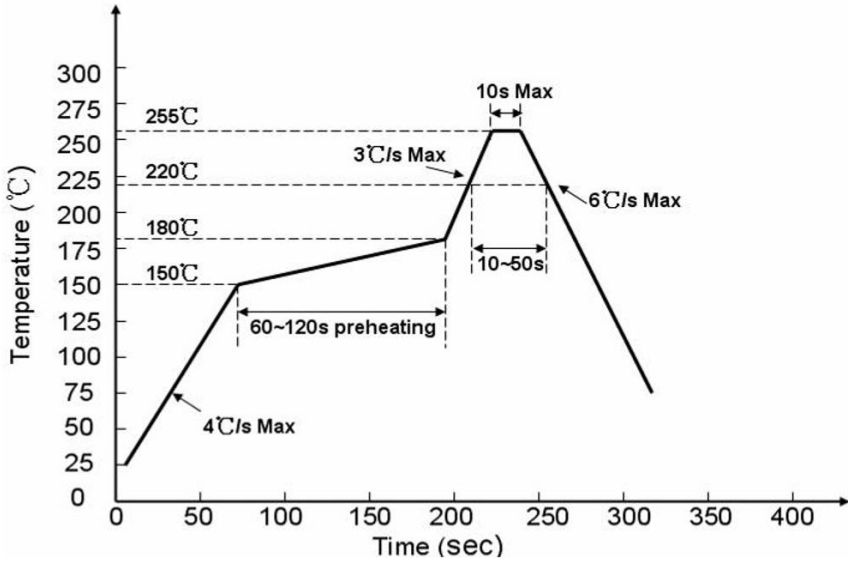
As shown in the figure below, the corresponding relationship between the magnetic field and the output, when the input magnetic field is 0mT (no magnetic field) and without considering the offset of hall chip, the output of the Hall sensor is 8192. When the south pole of the magnet is applied to the sensor, the output value of the sensor will decreases, and the decrease in the sensor output value changes linearly with the magnetic field strength until the sensor is saturated; correspondingly, when the north pole of the magnet is applied to the sensor, the sensor output value rises, and the increase in the sensor output value changes linearly with the S-type magnetic field strength until the sensor is saturated.



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13. Welding Requirements

13.1 SMT Reflow Soldering Instructions



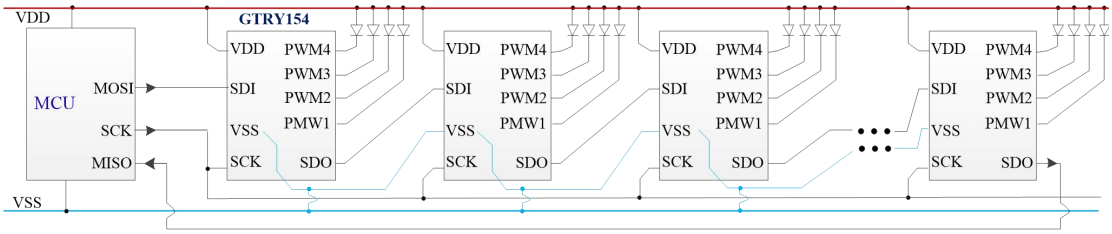
Note: It is recommended to use solder paste with high silver content and low particles, with an activity of 12 or more.

13.2 Hand Soldering Profile

The temperature of the iron should be lower than 350℃ and soldering within 3sec per solder-pad is to be observed.

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14. Reference Circuit



Note: Considering the reduction of EMI, it is strongly recommended to use 6~8 chains

15. Marking Information

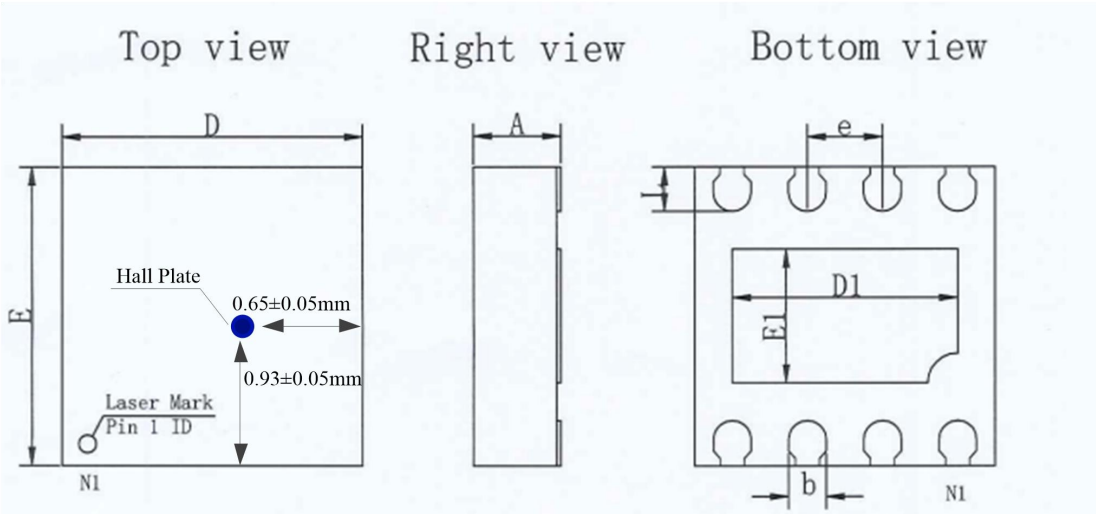
1) Marking:
① Line 1 is is product code : GTRY154
② Line 2 is the production batch number: YYWWXX. There may be hanges in this line, where YYWW is data code，YY is year，WW is week, and XX is the Serial Number

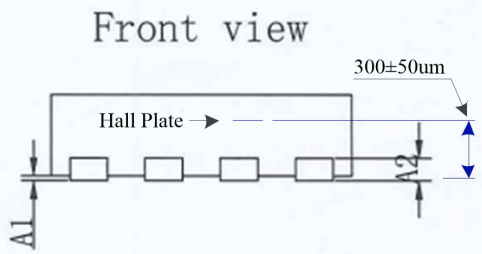
2) Examples of physical chips



Only for Customer Internal Reference

16. Package Outlines





COMMON DIMENSIONS
(UNITS OF MEASURE=MILLIMETER)

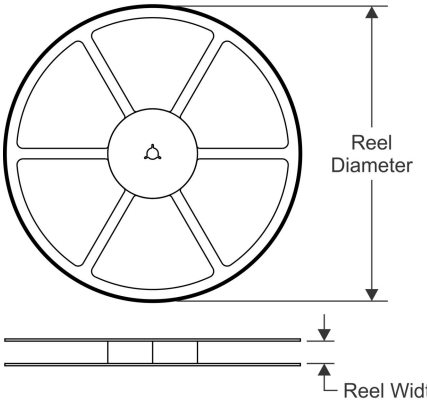
	SYMBOL	MIN	NOM	MAX
TOTAL THICKNESS	A	0.50	0.55	0.60
STAND OFF	A1	0.00	0.03	0.05
L/F THICKNESS	A2		0.152 REF	
LEAD WIDTH	b	0.20	0.25	0.30
BODY SIZE X	D	1.90	2.00	2.10
EP SIZE X	D1	1.40	1.50	1.60
BODY SIZE Y	E	1.90	2.00	2.10
EP SIZE Y	E1	0.79	0.89	0.99
LEAD PITCH	e		0.50 TYP	
LEAD LENGTH	L	0.25	0.30	0.35

Only for

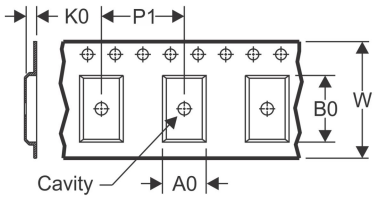
17. Package Materials Information

1. TAPE AND REEL INFORMATION

REEL DIMENSIONS

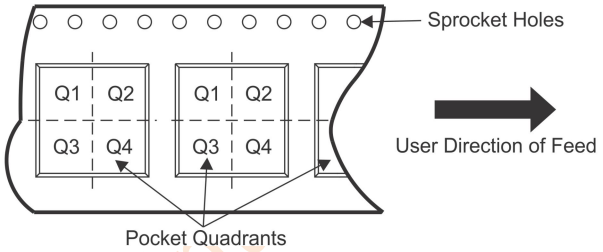


TAPE DIMENSIONS



A0	Dimension designed to accommodate the component width
B0	Dimension designed to accommodate the component length
K0	Dimension designed to accommodate the component thickness
W	Overall width of the carrier tape
P1	Pitch between successive cavity centers

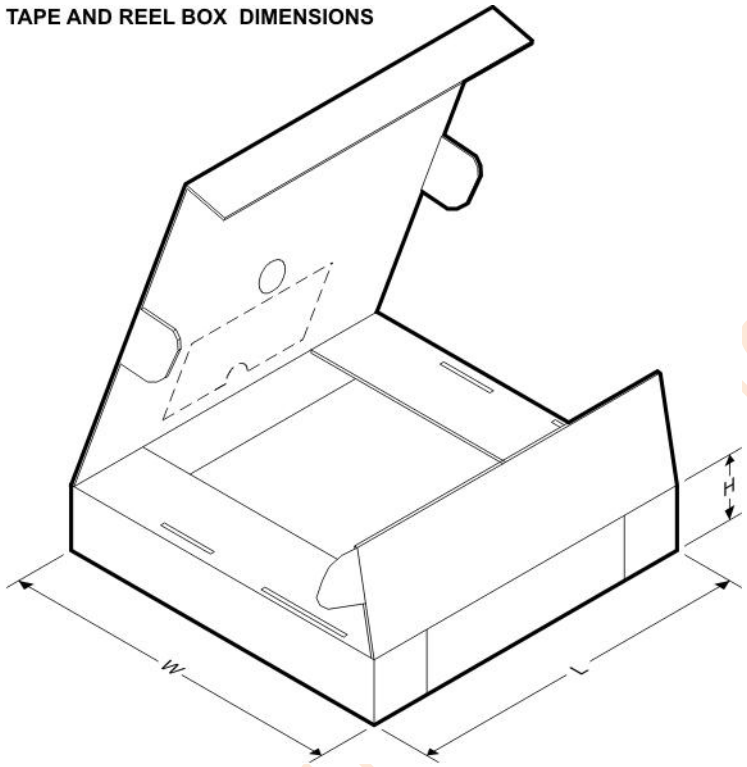
QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE



Device Name	Package Type	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
GTRY154	DFN2x2-8	8	4000	330	12.5	6.40	5.3	2.01	8	12	Q1

Only for Co... Inf...

2. TAPE AND REEL BOX DIMENSIONS



DeviceName	Package Type	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
GTRY154	DFN2x2-8	8	40000	357	340	50

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18. Summary of Changes

Rev	Paragraph(s) Changed	Description of Change(s)
V1.0		Initial version
V1.1	8	Update register description

Note: This document is an initial version, the specific parameter specifications are subject to the latest version of the manual,subject to change without notice

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